BUILDING API WITH NEST.JS + AI PROMPT

WHAT IS API, **WHY API** AND BUILD API WITH NEST.JS, MYSQ **AND AI PROMT**.





Cursor IDE

Development tools

TheLastCodeBender

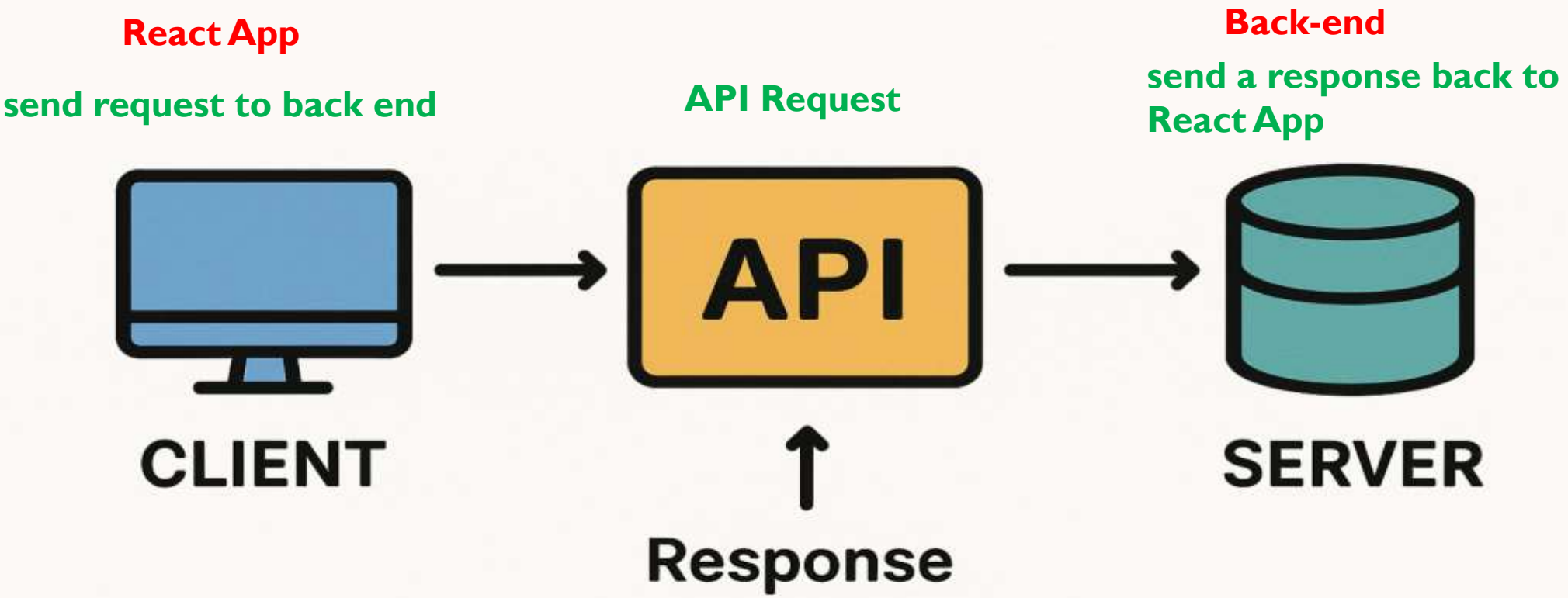
PART 1: WHAT IS API

Introduction to API

API stands for **Application Programming Interface**.

- It allows different software systems to **talk to each other**.
- In web development, APIs often send and receive **JSON data**.

Part 1:What is API



PART 1: WHAT IS NEST.JS

Nest.js is a **Node.js framework** for building server-side applications.
It uses **TypeScript by default** (but works with JavaScript too) and follows a **modular structure** (with controllers, services, and modules) to keep code clean and organized.

💡 Why We Need Nest.js

- Makes backend apps **easier to build and maintain**.
- Follows a **clear structure** (controllers, services, modules).
- Scales easily for **small apps or big enterprise apps**.

BY THE END OF THIS COURSE, YOU WILL BECOME A PROMPT/VIBE CODING ENGINEER BUILDING API WITH NEST.JS.

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PART 2: SETTING UP NEST.JS / INSTALLATIONS

Node.js download: https://nodejs.org/en

Postman download: https://www.postman.com/

Xampp download: https://www.apachefriends.org/download.html

PART 2: SETTING UP NEST.JS SERVER

Installation

Steps:

- **yarn global add @nestjs/cli**
- **Create a project: nest new my-project-name**
- **CD into my-project-name**
- **yarn start:dev**

PART 3: RESTFUL API ROUTES / NEST FOLDER STRUCTURE

In Nest.js, **RESTful routes** are a set of standard URL patterns that define how we **create, read, update, and delete (CRUD)** data.
They follow **REST (Representational State Transfer)** principles, making our APIs consistent and easy to understand.

Think of RESTful routes as a **set of rules** for how your server should respond to different types of HTTP requests (like **GET, POST, PUT, DELETE**).

PART 3: RESTFUL API ROUTES / NEST FOLDER STRUCTURE

✓ Definition

- **GET REQUEST** → Used to **read or fetch data** from the server.
- **POST REQUEST** → Used to **create new data** on the server.
- **PUT REQUEST** → Used to **update existing data** on the server.
- **DELETE REQUEST** → Used to **remove data** from the server.
- Together, they give us the **CRUD operations** (Create, Read, Update, Delete).

PART 4: CREATING RESTFUL ROUTES + AI PROMPT

Create Resource: run nest g resource resource-name

GET REQUEST → Used to **read or fetch data** from the server.

☐ AI **Prompt:**

Create a Get REQUEST that return the list of students using mock data.

PART 5: CREATING RESTFUL ROUTES + AI PROMPT

POST REQUEST → Used to **create new data** on the server.

☐ AI **Prompt:**

Create a POST REQUEST to store student information in mock data.

PART 6: CREATING RESTFUL ROUTES + AI PROMPT

PUT REQUEST → Used to **update existing data** on the server.

☐ AI **Prompt:**

Create a PUT REQUEST to update student information in mock data.

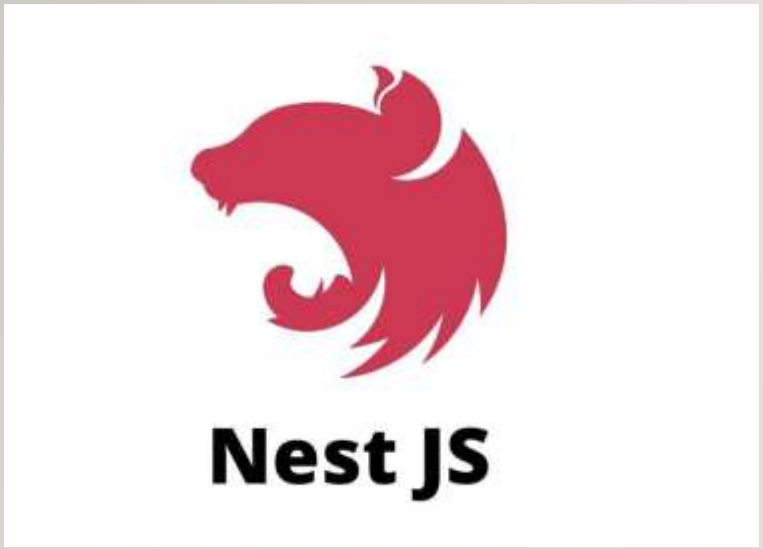
PART 7: CREATING RESTFUL ROUTES + AI PROMPT

DELETE REQUEST → Used to **remove data** from the server.

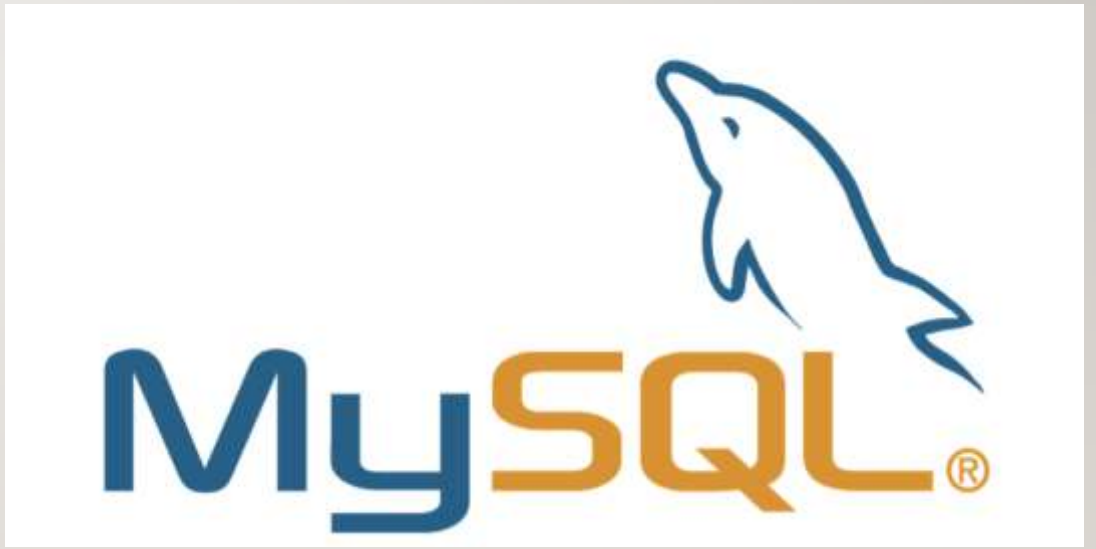
☐ AI **Prompt:**

Create a DELETE REQUEST to delete student information in mock data.

PART 8: CONNECTING TO MYSQL DB USING NEST.JS + AI PROMPT



Connection…. …….. →



PART 8: CONNECTING TO MYSQL DB USING NEST.JS + AI PROMPT

What is MySQL (and why use it)?

MySQL is a popular **relational database** that stores data in **tables** (rows & columns) and uses **SQL** to query/update it.

Why we use it:

- To permanently **save data/information** (e.g., users, cars, orders)
- **Fast, widely supported**, easy to host
- Works great with Nest.js via **TypeORM** or **Prisma**

Installation: yarn add @nestjs/typeorm typeorm mysql2 @nestjs/config

PART 9: REFACTOR RESTFUL ROUTES TO USE MYSQL DATABASE DATA + AI PROMPT

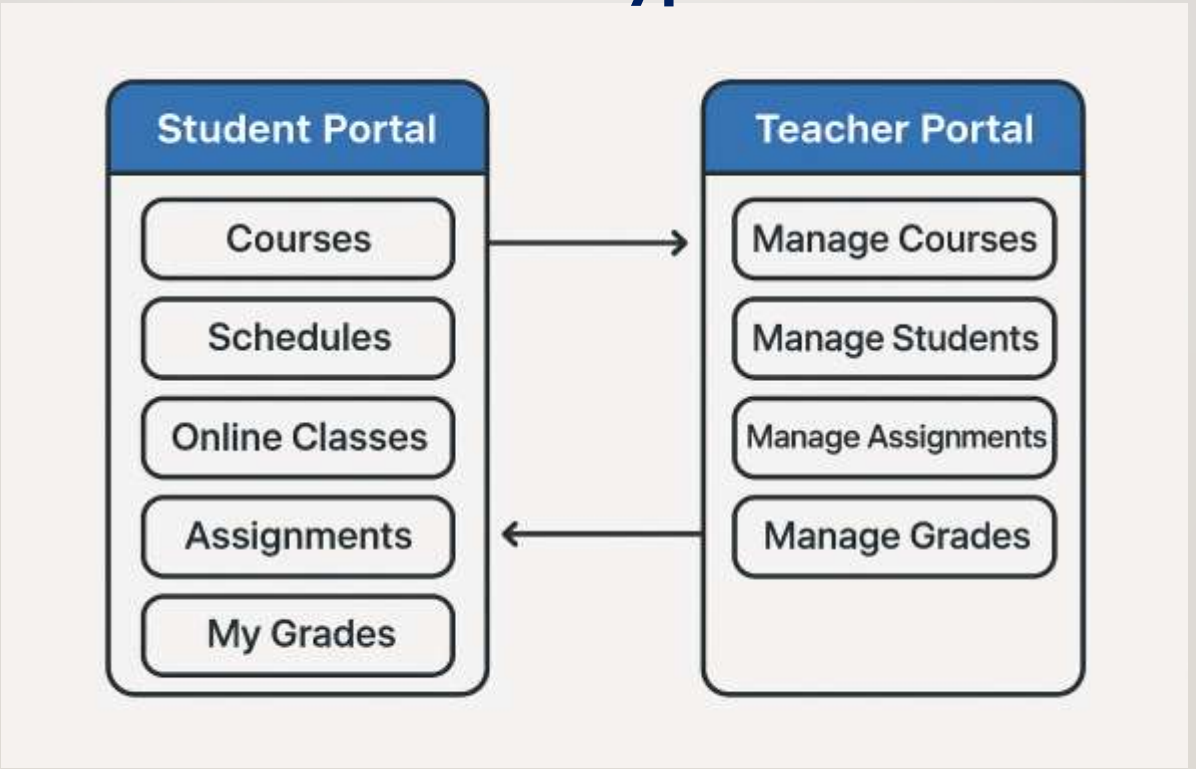
- ☐ AI **Prompt:**

Refactor car and student service to use data from MYSQL Database.

PART 10: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ **Remember our Golden Rule:**
One idea = One prompt.

User Types:



PART 10: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ AI **Prompts:**

Refactor registration page.

- - Let user select if they are a student or a teacher.
- - If they select student, they should be able to select their course, and Enrollment Term.
- - If they select teacher, they should sign up with their email and password and Full Name only.

PART 12: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ AI **Prompts:**

Connect the registration page with the backend.

- Create a user table with user type student and teacher.
- Redirect user to dashboard after registration successfully.

PART 13: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ AI **Prompts:**

Create Teacher dashboard.

- Update side bar navigation.
- If user is a student, show this menu: - courses, schedules, online-classes, assignments-exams, my-grades
- If user is a teacher, show this menu: Manage students, Manage courses, Manage assignments, Manage grades

Note: Always use yearn if you need to do any installation

PART 14: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ AI **Prompts:**

Connect Login page with the API. Redirect user to their dashboard base on the user type student or teacher

Note: Focus on Login page only and keep it simple.

PART 15: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ AI **Prompts:**

Connect Manage Students page with the API. A teacher can create, read, update or delete a student

Note: Focus on Students page only and keep it simple.

PART 16: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ AI **Prompts:**

Connect Manage Courses page with the API. A teacher can create, read, update or delete a course

Note: Focus on Courses page only and keep it simple.

PART 17: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ AI **Prompts:**

Connect Manage Assignments page with the API. A teacher can create, read, update or delete an assignment.

Note: Focus on Assignments page only and keep it simple.

PART 18: CONNECT OUR UNIVERSITY WEBSITE AND STUDENT PORTAL WITH BACKEND API + AI PROMPT

☐ AI **Prompts:**

Connect Manage Grades page with the API. A teacher can create, read, update or delete a grade.

Note: Focus on Grade page only and keep it simple.